

Qns	Answer	Marks
6(a)	upthrust is greater than weight (resultant force is) upwards	B1 B1
6(b)	At equilibrium, Upthrust = Weight $\rho A h g = M g \quad \text{---(1)}$ Taking downwards positive, at displacement x, Net force = mass $\times$ acceleration $M g - \text{Upthrust} = M a$ $M g - \rho V g = M a \quad \text{where } V \text{ is the volume of water displaced}$ $M g - \rho A (h + x) g = M a$ $M g - \rho A h g - \rho A x g = M a \quad \text{----(2)}$ Sub (1) into (2): $\rho A h g - \rho A h g - \rho A x g = M a$ $-\rho A x g = M a$ $a = -\left(\frac{A \rho g}{M}\right) x$ OR <u>Additional</u> upthrust = net force $\rho A x g = M a$ Since a and x are opposite in directions, $a = -\frac{A \rho g}{M} x$ Note: For “show that” question, the <u>first method</u> is preferred.	M1 M1 M1 (M1) (M1) M1
6(c)	A, ρ, g and M are constant, hence acceleration proportional to displacement negative sign indicates acceleration and displacement in opposite directions	B1 B1
6(d)(i)	$\omega = 2\pi / T$ $\omega = 2\pi / 1.3$ $= 4.8 \text{ rad s}^{-1}$	C1 A1
6(d)(ii)	$\omega^2 = A \rho g / M$ $4.8^2 = (4.5 \times 10^{-4} \times \rho \times 9.81) / 0.17$ $\rho = 900 \text{ kg m}^{-3}$	C1 C1 A1

Qns	Answer	Marks
6(d)(iii)1.	Total energy = <u>maximum</u> Kinetic energy	B1
	Max velocity $v_{\max} = \omega x_{\max}$	M1
	$= \frac{1}{2} m v_{\max}^2 = \frac{1}{2} m (\omega x_o)^2 = \frac{1}{2} m \omega^2 x_o^2$ algebra leading to final expression	M1
6(d)(iii)2.	$\frac{1}{2} (0.17) [(4.8)(0.20)]^2$ = 0.078 J	A1
6(d)(iv)	0.92^6	C1
	decrease in energy = $0.078 - (0.078 \times 0.92^6)$ = 30.7 mJ	A1
6(d)(v)	wave with same period (or slightly greater) AND peak height decreasing successively	B1 B1

